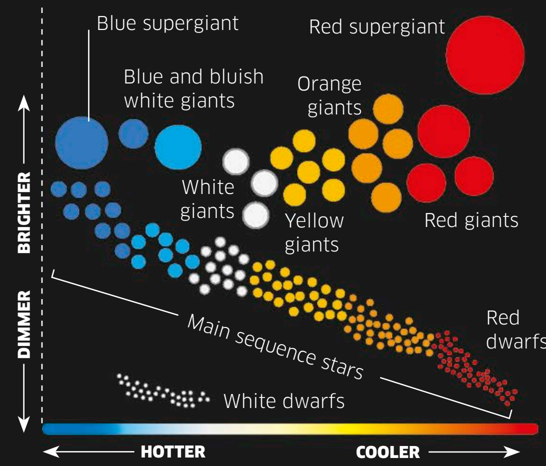
**If you sorted all the stars into piles, the biggest pile, by far, would be**
**red dwarfs—stars**
**with less than half of the sun's mass.**

## Star types

The Hertzsprung-Russell diagram is a graph that astronomers use to classify stars. It plots the brightness of stars against their temperature to reveal distinct groups of stars, such as red giants (dying stars) and main sequence stars (ordinary stars). Astronomers also classify stars by color, which relates to temperature. Red is the coolest color, seen in stars cooler than 6,000°F (3,500°C). Stars such as our sun are yellowish white and average around 10,000°F (6,000°C). The hottest stars are blue, with surface temperatures above 21,000°F (12,000°C).

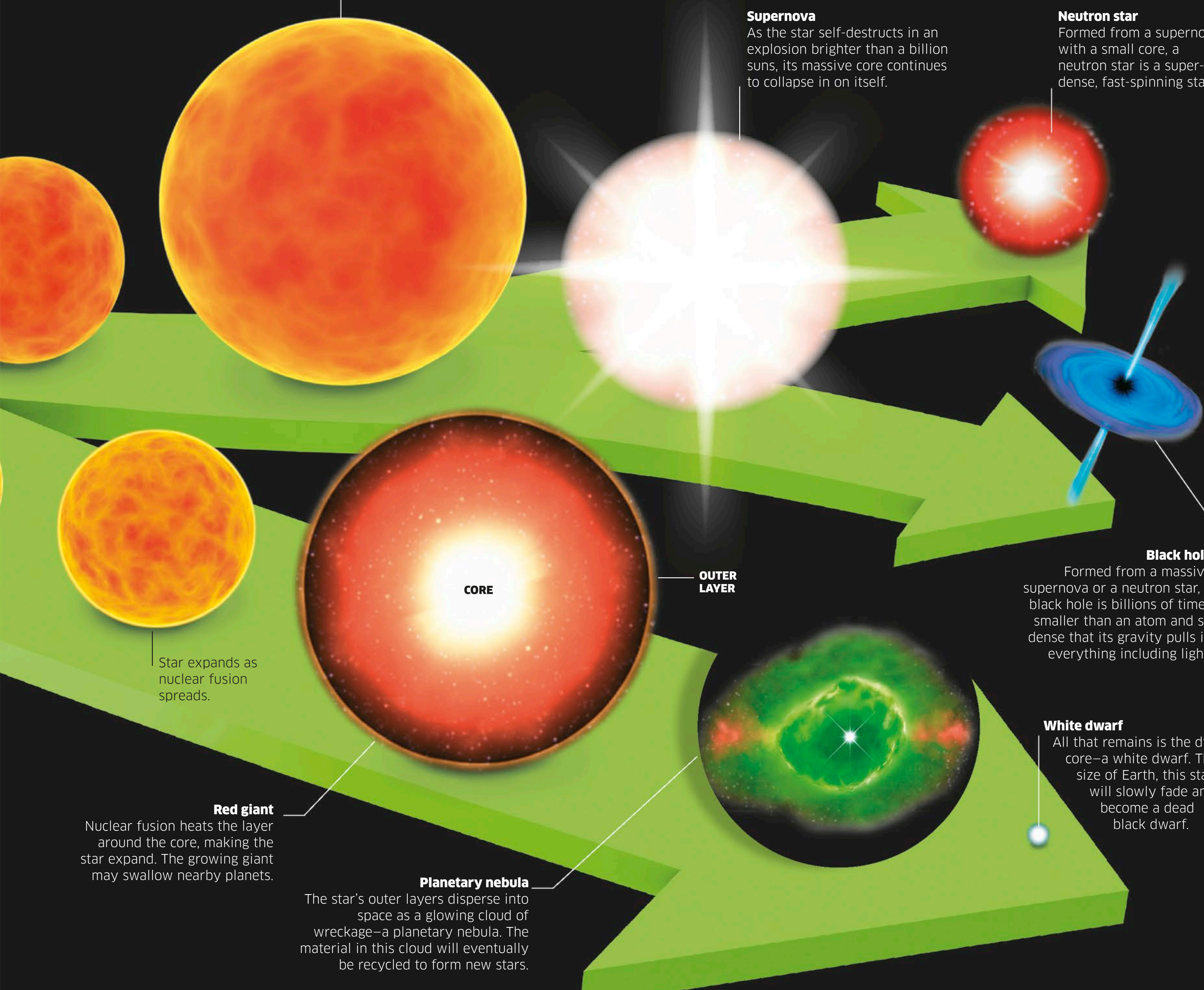



**Death of a massive star**
Stars more than eight times the mass of our sun will be hot enough to become supergiants. The heat and pressure in the core become so intense that nuclear fusion can fuse helium and larger atoms to create elements such as carbon or oxygen. As this happens, the stars swell into supergiants, which end their lives in dramatic explosions called supernovae. Smaller supergiants become neutron stars, but larger ones become black holes.

**Red supergiant**
Nuclear fusion carries on inside the core of the supergiant, forming heavy elements until the core turns into iron and the star collapses.

**Supernova**
As the star self-destructs in an explosion brighter than a billion suns, its massive core continues to collapse in on itself.

**Neutron star**
Formed from a supernova with a small core, a neutron star is a super-dense, fast-spinning star.




**Black hole**
Formed from a massive supernova or a neutron star, a black hole is billions of times smaller than an atom and so dense that its gravity pulls in everything including light.

Star expands as nuclear fusion spreads.

**Red giant**
Nuclear fusion heats the layer around the core, making the star expand. The growing giant may swallow nearby planets.

**Planetary nebula**
The star's outer layers disperse into space as a glowing cloud of wreckage—a planetary nebula. The material in this cloud will eventually be recycled to form new stars.

**White dwarf**
All that remains is the dying core—a white dwarf. The size of Earth, this star will slowly fade and become a dead black dwarf.